"PAPER_{0:D3}" -f [int]# PAPER #88 ■ Neutrino SED: UQFF Emission Model

Author: Daniel T. Murphy

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1. Physical Origin of Neutrinos from BH Systems

Three UQFF neutrino production channels:

Channel	UQFF Term	Energy Range	Flux Contribution
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Corona pp/p?	Ug4 AGN feedback	1 TeV ■ 10 PeV	Dominant
TRZ vacuum	f_TRZ = 0.01	0.1■100 PeV	1% enhancement

For stellar and SMBH systems, **Hawking neutrinos are negligible** (T*H ~ 10?■4 K ? E*? ~ k*B T*H << 1 eV). The interesting band is TeV■PeV cosmic neutrinos from AGN corona interactions.

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2. The UQFF Neutrino SED

2.1 Standard Model Corona Prediction

For Sgr A* in active state with corona luminosity L*corona = 10?*■ *L*Edd:

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\Phi_\nu^{\rm SM}(E_\nu) = \Phi_0 \left(\frac{E_\nu}{\rm TeV}\right)^{-\gamma}
e^{-E_\nu/E_{\rm cut}}
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With ? = 2.2, E_cut = 5 PeV.

2.2 UQFF Enhancement via TRZ

The Toroidal Resonance Zone factor f_TRZ = 0.01 enhances corona pair production:

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\Phi_\nu^{\rm UQFF}(E_\nu) = \Phi_\nu^{\rm SM}(E_\nu) \times (1 + f_{\rm TRZ}) = 1.01
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2.3 Ug4-Mediated Enhancement

The Ug4 energy density (3.352941 ■ 10■■ J/m■ baseline) additionally contributes to pion production:

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\Delta \Phi_\nu^{Ug4}(E_\nu) = f_{Ug4} \cdot \Phi_\nu^{\rm SM} \cdot
\frac{U_{g4}}{U_{g4,\rm ref}}
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For current Sgr A* quiescent state (A*AGN = 1, e*$^{-?t}$ *near-unity): f*Ug4 << f_TRZ.

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Standard model predictions: ?*e : ?*■ : ?_t = 1:2:0 at production ? 1:1:1 after oscillation.

UQFF modification via 26D channel mixing (Batch 21):

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(\nu_e : \nu_\mu : \nu_\tau)_{\rm UQFF} = (1 : 1 : 1) \times (1 + 0.001 \, f_{\rm TRZ})
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The 26D mixing is degenerate in flavor at the detector level ■ **no measurable deviation from 1:1:1** at IceCube sensitivity. This prediction is consistent with IceCube observations showing approximate (but not exact) flavor democracy.

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From MAIN*1*CoAnQi.cpp Batch 21:

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double TRZ_factor = 1.0 + f_TRZ; // = 1.01
double Ug4_factor = 1.0 + f_Ug4 * Ug4_current / Ug4_ref;
return SM_flux * TRZ_factor * Ug4_factor;
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5. validate_phase3.py Validation

`validate_phase3.py` performs phase-3 cross-validation of Batch 21 outputs against `uqff_validation_test.py`:

Neutrino SED test: UQFF flux = 2s deviation from IceCube diffuse background ? **PASS**

Multi-flavor test: (1:1:1) ratio ? **PASS**

TRZ enhancement: ? = 1.0% above SM ? **PASS**

Ug4 quiescent: negligible Ug4 contribution at A_AGN = 1 ? **PASS**

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6. IceCube-Gen2 Detectability

Current IceCube sensitivity: F_3s ■ 10?■■ TeV cm?■ s?■ sr?■ for Sgr A* point source.

UQFF prediction: ? = 0.3% excess above SM corona prediction.

Not detectable by IceCube-Gen2 (~10-year run). However, an exaggerated AGN-active Sgr A* state (A_AGN >> 10) would push Ug4 enhancement to ~5% ■ potentially within 3s of Gen2.

Summary

Prediction	UQFF Value	Standard Model	UQFF Signature
TRZ enhancement	+1.0%	0%	UQFF-specific
Flavor ratio	1:1:1 (■1.001)	1:1:1	Unmeasurable
Quiescent Ug4	negligible	None	Consistent
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Phase 3 validation	4/4 PASS	N/A	Verified

Source: MAIN1CoAnQi.cpp Batch 21 (NeutrinoSEDModule) | validatephase3.py | fTRZ = 0.01 | IceCube constraints .Groups[1].Value "PAPER$_{0:D3}$" -f $n

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UQFF computed: UQFF-modified neutrino mass shift delta*mnu* = 2.85e-4 ■ v■/*MN; for M*N = 1.0e14 GeV: delta*mnu* = 5.1e-16 GeV.

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references
> the production physics constants and master equations to enable reproducibility
> against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	5.0 × 10⁻■ day^{-1}	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k■	1.5	Ug1 DPM-dipole coupling
k■	1.2	Ug2 outer-bubble charge coupling
k■	1.8	Ug3 string-rotation coupling
k■	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
E_react(0)	10■■ J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

```
F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4}■igl[\lambda_i
\cdot U_i(r,t) \cdot E_{\mathrm{react}}■igr]
```

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
–Σλ■·U■·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420): λ■=10^{-10}, λ■=10^{-12}, λ■=10^{-11}, λ■=10^{-13} (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

```
U_m^{\mathrm{full}} = U_m^{\mathrm{base}} imes ■igl(1 + 10^{13}\,\Theta(
ho_{SCm} -
ho_c)■igr) imes ■igl(1 + A_q\cos(\Delta\omega\,t)■igr)
```

Symbol	Value	Description
ρ_c	10^{1}■ kg/m^{3}	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434\cdot365.25)$ rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta\_i \times$ Ubi	Expanding nebulae, stellar winds
Superconductive	Um $\times$ $(1+10^{13}\cdot$f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, *and* `CondensedPhysics2.py`.